

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 676 Data Mining — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Leskovec, Rajaraman & Ullman — Mining of Massive Datasets (free PDF)	The primary text, free in full from the authors. Modules 02, 04, 06, 07, and 09 follow it closely, and its treatment of locality-sensitive hashing is the best available.
Tan, Steinbach, Karpatne & Kumar — Introduction to Data Mining, 2nd edition	The broader companion, with free chapters. Stronger on Modules 03, 05, and 08, and on the evaluation material generally.
Aggarwal — Outlier Analysis, 2nd edition	Module 08 in depth — and honest about how badly anomaly detection is usually evaluated. Library copy.
Benjamini & Hochberg, and the replication literature (free)	Module 11's methods and its motivation. The FDR paper itself, read alongside the reproducibility work that made it urgent.
Dwork & Roth — The Algorithmic Foundations of Differential Privacy (free PDF)	Module 12's formal half, free in full. Read the first three chapters; the definition is what matters for this course.
Stanford CS246 — Mining Massive Data Sets (free)	The course this one is shaped against , with free lectures and assignments by the authors of the primary text.

Module-by-module map

Module 01 — Two Problems, Not One

Resource	What it covers for this module
Mining of Massive Datasets, chapter 1 (free)	§1 through §3 — and its section on 'bonferroni's principle' is this course's §3, stated by the authors with a memorable example.
Ioannidis — Why Most Published Research Findings Are False (free)	§3's argument with the arithmetic — short, and it changed how several fields report results.
Simmons, Nelson & Simonsohn — False-Positive Psychology (free)	§4's motivation — how ordinary analytic flexibility produces significant findings from nothing.
Stanford CS246, lecture 1 (free)	§2's scale thresholds , with the cost model argued concretely.

Module 02 — Similar Items at Scale

Resource	What it covers for this module
Mining of Massive Datasets, chapter 3 (free)	The whole module, and the best treatment in print — including the parameter selection worked through numerically.
Broder — On the resemblance and containment of documents (free)	§2 in the original — minhashing as it was invented, for exactly this problem.
Indyk & Motwani — Approximate nearest neighbors (free)	§4's general framework — where locality-sensitive hashing is defined properly.
Charikar — Similarity estimation from rounding algorithms (free)	§4's first row — the random hyperplane family and its collision probability.

Module 03 — Frequent Patterns

Resource	What it covers for this module
Tan et al. — Introduction to Data Mining, chapters 5 and 6	The whole module , with the interestingness measures treated more carefully than elsewhere.
Mining of Massive Datasets, chapter 6 (free)	§2's algorithms with the memory analysis — and PCY is explained better here than anywhere.
Agrawal & Srikant — Fast Algorithms for Mining Association Rules (free)	§2's first algorithm in the original , with the closure property stated as the key insight.
Han et al. — Mining Frequent Patterns without Candidate Generation (free)	§2's second algorithm — FP-growth, and the pass-count argument for it.

Module 04 — Clustering at Scale

Resource	What it covers for this module
Mining of Massive Datasets, chapter 7 (free)	§2's scale algorithms — BFR and CURE, with the assumptions stated clearly.

Resource	What it covers for this module
Tan et al., chapters 7 and 8	§1 and §3 — and the cluster validity chapter is the best available treatment of the evaluation problem.
von Luxburg — A Tutorial on Spectral Clustering (free)	§1's fourth row, properly — and the connection to graph cuts, which Module 07 uses.
Ben-David, von Luxburg & Pál — A Sober Look at Clustering Stability (free)	§3's stability measure, examined critically — including when it does not work.

Module 05 — Dimensionality Reduction

Resource	What it covers for this module
Tan et al., the dimensionality reduction appendix	§1 and §2 , with the curse of dimensionality developed carefully.
Mining of Massive Datasets, chapter 11 (free)	§2 and §3 — SVD, CUR, and the scale considerations, free in full.
Wattenberg, Viégas & Johnson — How to Use t-SNE Effectively (free)	§4's caution, demonstrated interactively — the single best corrective to the common misreading.
Achlioptas — Database-friendly random projections (free)	§3 with a projection matrix you can compute cheaply — the practical version of the lemma.

Module 06 — Streams and Sketches

Resource	What it covers for this module
Mining of Massive Datasets, chapter 4 (free)	The whole module — the streaming model, the sketches, and the sampling, with the analyses.
Cormode & Muthukrishnan — An improved data stream summary (free)	§3's Count-Min in the original , with the bound derived — short and readable.
Flajolet et al. — HyperLogLog (free)	§3's fourth row — and the analysis is a genuinely elegant piece of work.
Cormode & Yi — Small Summaries for Big Data	§3 and §4 comprehensively — including the lower bounds and the windowed variants. Library copy.

Module 07 — Graph Mining

Resource	What it covers for this module
Mining of Massive Datasets, chapters 5 and 10 (free)	§2's PageRank and §3's community detection , with the scale considerations throughout.
Easley & Kleinberg — Networks, Crowds, and Markets (free PDF)	§1 and §2 in their social context , free in full — and the clearest treatment of why these properties arise.

Resource	What it covers for this module
Fortunato & Barthélemy — Resolution limit in community detection (free)	§3's limitation, proved — short, and it should be read by anybody using modularity.
Malewicz et al. — Pregel (free)	§4's programming model — think-like-a-vertex, and why it fits this class of algorithm.

Module 08 — Anomaly Detection

Resource	What it covers for this module
Aggarwal — Outlier Analysis, 2nd edition	The whole module — and its chapter on evaluation is the most honest treatment available. Library copy.
Liu, Ting & Zhou — Isolation Forest (free)	§2's default method in the original — short, and the intuition is well explained.
Breunig et al. — LOF: Identifying Density-Based Local Outliers (free)	§2's density row — and the varying-density problem it was designed to solve.
Campos et al. — On the evaluation of unsupervised outlier detection (free)	§4's problem, measured — a systematic study showing how much the evaluation choices change the conclusions.

Module 09 — Recommendation

Resource	What it covers for this module
Mining of Massive Datasets, chapter 9 (free)	§1 and §2 , with the collaborative filtering and latent factor material done cleanly.
Hu, Koren & Volinsky — Collaborative Filtering for Implicit Feedback (free)	§2's callout in the original — the confidence weighting approach, and why the problem differs.
Dacrema, Cremonesi & Jannach — Are We Really Making Much Progress? (free)	§4's replication finding — the paper that forced the field to tune its baselines. Read it before Module 13.
Chaney, Stewart & Engelhardt — How algorithmic confounding biases recommendation (free)	§3's feedback loop, simulated — what happens to the data over time under a deployed recommender.

Module 10 — Scale and Systems

Resource	What it covers for this module
Mining of Massive Datasets, chapter 2 (free)	§1's models, with the communication cost analysis — and the replication-rate framework for designing MapReduce algorithms.
Zaharia et al. — Resilient Distributed Datasets (free)	§1's second model in the original — and the lineage recovery argument, which is the interesting part.

Resource	What it covers for this module
McSherry, Isard & Murray — Scalability! But at what COST? (free)	§3's argument, measured — the paper every student of this material should read before provisioning a cluster.
Spark's tuning and SQL performance documentation (free)	§2 and §4 practically — the skew handling and the partitioning guidance, from the implementation.

Module 11 — The Pattern You Found

Resource	What it covers for this module
Benjamini & Hochberg — Controlling the False Discovery Rate (free)	§2's third row in the original — and the argument for why the FDR is the right target when you are screening.
Gelman & Loken — The Garden of Forking Paths (free)	§1's hardest point — that an implicit hypothesis count exists even without explicit multiple testing.
Ioannidis — Why Most Published Research Findings Are False (free)	§1 and §4's motivation , with the arithmetic — reread it now that you have the methods.
Sullivan & Feinn — Using Effect Size (free)	§3's recommendation, practically — which effect size measures to use and how to report them.

Module 12 — Privacy

Resource	What it covers for this module
Dwork & Roth — The Algorithmic Foundations of Differential Privacy (free PDF)	§3's first three chapters — the definition, the basic mechanisms, and composition, which is all this course needs.
Narayanan & Shmatikov — Robust De-anonymization of Large Sparse Datasets (free)	§1's mechanism, demonstrated — and the clearest account of why auxiliary information is the problem.
Machanavajjhala et al., and Li et al. — l-diversity and t-closeness (free)	§2's patch sequence in the originals — read them as a sequence, which is the lesson.
Dinur & Nissim — Revealing Information while Preserving Privacy (free)	§4's reconstruction result — the theorem that aggregates determine the records.

Module 13 — Claiming a Pattern Honestly

Resource	What it covers for this module
Mining of Massive Datasets, chapter 1 (free)	§1's position, from the authors — reread the bonferroni's-principle section now that you have Module 11.
Ioannidis, and the reproducibility literature (free)	§1 and §4's motivation — the third reading, and it reads differently each time.

Resource	What it covers for this module
Dacrema et al. — Are We Really Making Much Progress? (free)	§2's baseline requirement, demonstrated — and Module 09 §4's finding, which generalises.
Leek & Peng — Statistics: P values are just the tip of the iceberg (free)	§4's ordering — why the design decisions matter more than the statistical test.